

TrES-1b

R-2132

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110523 · created 2026-08-11T23:32:35+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455704.79594 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.017
Transit depth (Rp/Rs)^2	1.46 +/- 0.41 [%]
Orbital Inclination (inc)	86.44 +/- 0.76
Airmass coefficient 1 (a1)	1.023 +/- 0.012
Airmass coefficient 2 (a2)	-0.0133 +/- 0.0039
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	#3 - [460, 125]
Optimal Aperture	4.63
Optimal Annulus	15.98
Transit Duration (day)	0.082 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.017 vs 0.1358 (deviation 0.87σ)
Duration fit vs published	0.082 d vs 0.1042 d (deviation 2.02σ)
Mid-time O–C	0.6 min
Depth SNR	7.1
Residual scatter	1.17 %

Light curve

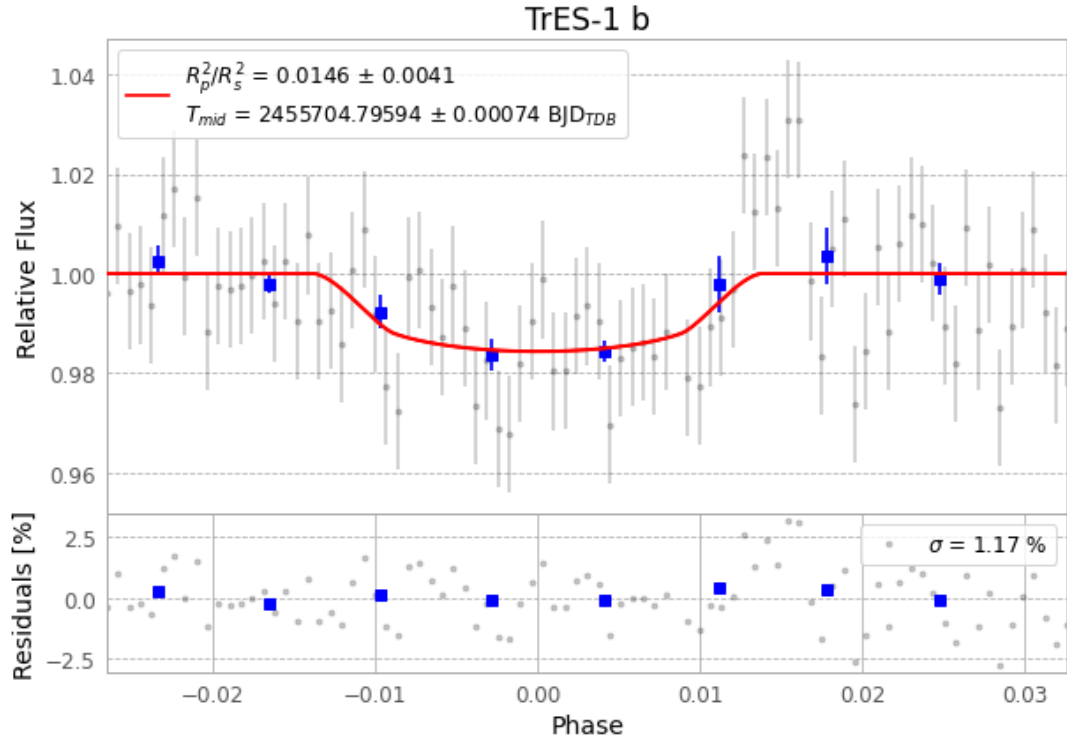


Figure 1 — FinalLightCurve_TrES-1 b_2011-05-22.png (SHA-256 da0a0f3a9c8a8dc2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e2e6e1080ba00937...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

87 FITS frames (57 MB), TRES-1110523050612.FITS → TRES-1110523092413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110523.log (43006aaf114a...), FinalLightCurve_TrES-1 b_2011-05-22.png (da0a0f3a9c8a...), FinalLightCurve_TrES-1 b_2011-05-22.csv (d20aacc9387c...)

Compute resources

Wall time 182.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:32Z.